

Astrocyte tissue modeling and the impact of ischemic stroke

Carlos Danilo M. Regis, Felipe R. G. Ribeiro, Rafael D. de Sousa and Smith L. da Silva

ABSTRACT

Poster: The brain damage caused by a stroke and how the communication between non-neuronal cells is affected is not fully understood. This paper aims to investigate these effects using a model of calcium signaling considering that a region of the tissue is affected by ischemia, one cell acts as a transmitter and comparing the gain on different receptor positions, different types of connection and whether the ischemia is present. The differences in the gain values between the simulations with stroke and without stroke were up to 12.7dB and 10.15dB.

CCS CONCEPTS

• **Computer systems organization** → **Embedded systems**; *Redundancy*; *Robotics*; • **Networks** → *Network reliability*; • **Computing methodologies**;

KEYWORDS

Stroke, astroglial network, molecular communication, computational model

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1 INTRODUCTION

According to the World Health Organization, it is estimated that around 50 million people suffer from some form of dementia, of which 60% of this total are affected by Alzheimer's Disease while the second leading cause of dementia is stroke, usually classified as ischemic or hemorrhagic [5].

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Thus, this work aims to evaluate the effects of a stroke in the ion-based communication between astrocytes, similar to what has been done for Alzheimer's Disease. The theoretical framework for this work consists of the calcium ions signaling and the effect of the stroke on this signaling and is presented in the second and third sections, respectively.

2 THE INFLUENCE OF A STROKE IN THE SIGNALING PROCESSES

The intercellular communication between astrocytes can be inositol triphosphate (IP_3) based or based on neurotransmitters, occurring between astrocytes that communicate between themselves through chemical synapses called gap-junctions. To simulate these chemical reactions involving Ca^{2+} ions, the Gillespie stochastic direct method was used in previous related works [2] to create an application similar to the one presented herein.

Normal levels in the extracellular Ca^{2+} concentration are a necessary condition for most of the toxins to lead to cell death, as well as massive shifts of this ion from the extracellular environment into the cell [3]. On the other hand, an increase in the Ca^{2+} intracellular levels occurs with a decrease in the adenosine triphosphate (ATP), essential for the ion pumps to maintain homeostasis [1, 3]. This decrease is due to the lack of glucose and the lack of oxygen. This inhibition in the cell membrane transport, can lead to neuron deaths induced by ischemia [4].

The mathematical model for the type of communication used in the current work consists of the relationship between the intracellular concentration of Ca^{2+} ($[Ca^{2+}]_i$), the concentration flow in the endoplasmic reticulum (ER) (E_a), and the variation in the IP_3 concentration (I_a), as shown in Equation 1. The factor $\kappa_s(Ca_s)$, was added in the current work to model the occurrence or non-occurrence of the stroke.

$$\frac{d[Ca^{2+}]_i}{dt} = \sigma_0 - \kappa_0[Ca^{2+}]_i + \sigma_1 - \sigma_2 + \kappa_f(E_a - [Ca^{2+}]_i) + \kappa_s(Ca_s) \quad (1)$$

The communication process consists of a transmitter, the channel, and the receptor. The transmitter is the cell responsible for the transmission of Ca^{2+} signals, based on a structure with an infinite amount of calcium. In this scenario, the transmitter is at the center of the tissue [2]. The channel is an n -dimensional network of astrocytes connected by one of the

topologies: regular degree and link radius. In the simulated tissue, each astrocyte has an initial calcium concentration that is controlled by natural cell propagation methods [2]. In the channel, a structure of cells with high static concentrations of calcium was inserted to represent the cells affected by the stroke. The receptor is an astrocyte a few cells away from the transmitter. Once the concentration reaches certain levels, this receptor is activated and starts to receive new molecules [2].

In this model, a region of the tissue has the characteristics of the core of an ischemic stroke, with constant Ca^{2+} concentration values [6]. In this case, the differential equation of calcium variations, presented by [2], needs to be revised by the addition of a $\kappa_s(Ca_s)$ factor, as shown in Equation 1. $\kappa_s > 0$ if the tissue is affected by the stroke, and Ca_s are the calcium concentration values in the stroke region.

3 EVALUATION AND PERFORMANCE

The proposed model implemented using an stochastic solver similar to the one used by [2] contains the following characteristics: the simulation consists of the first 800 seconds after the lesion; two different topologies are tested separately; the transmissions begin at the same time of the stroke event as the middle cell of studied tissue; the receiver is positioned in four different places to evaluate the interference caused by calcium stores on stroke cells. Considering each topology and receptor position, ten simulations were made and the average gain values of these simulations are presented in Table 1.

Table 1: Transmission gain for simulation for different receiver positions and different stroke sizes.

Distance	Topology	Gain [dB]			
		1	2	3	4
Without stroke	Regular Degree	-7.42	-14.83	-21.26	-24.50
With stroke S=1	Regular Degree	-7.46	-14.84	-21.18	-24.20
With stroke S=2	Regular Degree	-7.50	-14.63	-19.33	-20.58
With stroke S=3	Regular Degree	-7.08	-10.85	-11.71	-11.80
Without stroke	Link Radius 2	-9.10	-9.44	-14.87	-16.52
With stroke S=1	Link Radius 2	-9.03	-9.26	-14.25	-15.06
With stroke S=2	Link Radius 2	-8.19	-8.00	-9.75	-9.72
With stroke S=3	Link Radius 2	-6.62	-5.63	-6.59	-6.37

For the regular degree topology, the stroke effects are perceptible on the communications process if cells are sufficiently close to these damaged cells as noticed when the gains are analyzed together in each test, as summarized in Table 1, where the size of each cell S is the measurement of the side of a cubic tissue affected by stroke. On the other hand, for the link radius topology with a distance of two cells, the effects of cerebral dysfunctions are lesser than expected.

In this case, the transmitter contributions are more expressive. Figure 1 shows these contributions graphically, which are visible on upper locations of the graph the fake colors are at the same colors of the receiver position. Similarly, the regular degree results are summarized in Table 1.

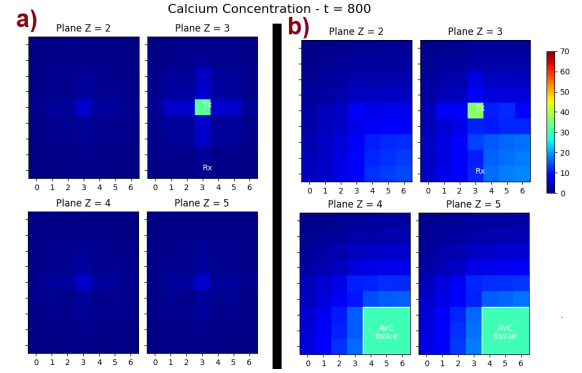


Figure 1: End state of the cellular communication simulation with a stochastic solver based on the one proposed by [2] using link radius 2 topology (a) Tissue without stroke; (b) Tissue with stroke.

4 CONCLUSIONS

The current work only assesses the effects of the Ca^{2+} and only in astrocytes, since they are a relevant type of signaling in the organism. Furthermore, it is demonstrated that the effects of abnormal calcium stores are key events in the degenerative process.

For the receiver with distance four, the stroke tissue area 3 and the tissue without stroke have a gain difference of 12.7dB and 10.15dB for the regular degree and link radius topologies, respectively.

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